Reproducibility Sheet: Supervised Learning Report

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Course: CS7641 Machine Learning - Spring 2026

(a) READ-ONLY Overleaf Link

Overleaf Project Link:
https://www.overleaf.com/read/snswsdqjrxxd#d205c3

(b) GitHub Commit Hash

8f397be2e4b6744afeb0e7317ab0708555372a2f

Repository:
https://github.gatech.edu/gt-omscs-ml/cs-7641-2026-spring-urafi3

(c) Exact Run Instructions

Folder Structure

- data/ – contains the raw CSV files (adult.csv, wine.csv).
- outputs/ – generated figures and JSON result files.
- src/ – Python modules for data loading, modeling, plotting, and utilities.
  - 'init.py'
  - 'data_loader.py'
  - inspect_data.py – quick data inspection script.
  - models.py -
  - paths.py – central configuration of paths and random seed.
  - plotting.py –
  - run_analysis.py – main script to run the entire experiment.
  - utils.py -
- requirements.txt – list of required Python packages.

Setup

1. Place the dataset files in the data/ folder.
2. Install dependencies:

pip install -r requirements.txt

3. In the parent directory (of the src/ folder), type: **python -m src.run_analysis**

(d) EDA Summaries

Adult Income Dataset

The dataset contains 48,842 instances after cleaning, with 14 features split between 6 numerical and 8 categorical variables. The task is binary classification: predicting whether income is ≤$50K or >$50K. There's a moderate class imbalance with 75% earning ≤$50K (36,631) and 25% earning >$50K (12,211), giving roughly a 3:1 ratio.

For numerical features, age ranges from 17-90 (mean ~38.6) and hours-per-week ranges 1-99 (mean ~40.4). Education-num captures years of education (1-16). Both capital-gain (max 99,999) and capital-loss (max 4,356) are heavily right-skewed with most values at zero, indicating few people have significant investment income. The fnlwgt (final weight) variable has wide-ranging values.

Categorical features include workclass (8 levels: Private, Self-employed, Government, etc.), education (16 levels from Preschool to Doctorate), marital-status (7 categories), occupation (14 types including Tech-support, Sales, Exec-managerial), relationship (6 types), race (5 categories), sex (Male/Female), and native-country (United-States plus 40+ others).

Missing values appear in workclass, occupation, and native-country. I handled these by creating a 'missing' category for categoricals and using median imputation for numericals.

Wine Quality Dataset

This dataset has 6,497 wine samples with 12 numerical features plus a binary wine type indicator (red=0, white=1). The target is quality rating from 3-9, though the distribution is highly imbalanced: most wines cluster around quality 5 (33%) and 6 (44%), with quality 7 at 17%. The extremes are rare—only 30 samples rated 3, and just 5 rated 9.

The 11 physicochemical features include acidity measures (fixed: 3.8-15.9, volatile: 0.08-1.58, citric: 0-1.66), residual sugar (0.6-65.8, right-skewed), chlorides (0.009-0.611), sulfur dioxide levels (free: 1-289, total: 6-440), density (0.987-1.039), pH (2.72-4.01), sulphates (0.22-2.0), and alcohol content (8.0-14.9%). Most features have reasonable ranges except residual sugar, which shows high variability.

There are no missing values in this dataset.